

Weekly Notice 3

Exercises, week 3:

1. Exercise 27. Conditional variances are defined in equation (1) below. Try to make use of Theorem D.10 as much as possible. However, for the first question, you will likely need to apply The Useful Rule.
2. Exercise 37. In (2) you may want to use Corollary 4.22, which will be covered in the lecture on Monday.
3. Exercise 38. You can verify the Markov property by appealing to Example 4.9, thus avoiding proof by hand-waving. To do this, define and interpret $(\epsilon_n)_{n \geq 1}$, and explain how they determine the "number of throws since the last six" and the "total number of pips".
4. Exercise 34.
5. Exercise 26.
6. Exercise 31.

Remark: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, Z be a square integrable random variable, and $\mathcal{G}$ be a σ -field with $\mathcal{G} \subseteq \mathcal{F}$. In this case $\mathbb{E}[Z | \mathcal{G}]$ is square integrable as well and $\text{Var}(Z | \mathcal{G})$ is defined as

$$\text{Var}(Z | \mathcal{G}) = \mathbb{E} \left[\left(Z - \mathbb{E}[Z | \mathcal{G}] \right)^2 \middle| \mathcal{G} \right]. \quad (1)$$

An alternative way of calculating this conditional variance is

$$\text{Var}(Z | \mathcal{G}) = \mathbb{E}[Z^2 | \mathcal{G}] - (\mathbb{E}[Z | \mathcal{G}])^2. \quad (2)$$

Indeed,

$$\begin{aligned} & \mathbb{E} \left[\left(Z - \mathbb{E}[Z | \mathcal{G}] \right)^2 \middle| \mathcal{G} \right] \\ &= \mathbb{E} \left[Z^2 + (\mathbb{E}[Z | \mathcal{G}])^2 - 2Z \cdot \mathbb{E}[Z | \mathcal{G}] \middle| \mathcal{G} \right] \\ &= \mathbb{E}[Z^2 | \mathcal{G}] + (\mathbb{E}[Z | \mathcal{G}])^2 - 2\mathbb{E}[Z | \mathcal{G}] \cdot \mathbb{E}[Z | \mathcal{G}] \\ &= \mathbb{E}[Z^2 | \mathcal{G}] - (\mathbb{E}[Z | \mathcal{G}])^2, \end{aligned}$$

where we have used that $\mathbb{E}[Z | \mathcal{G}]$ is $\mathcal{G}$ -measurable.